ĐẠI HỌC QUỐC GIA HÀ NỘI
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VNU-INTERNATIONAL SCHOOL

Project Monitoring and Control

INS3044: IT Project Management

Define execution phase stages.

Understand the control cycle.

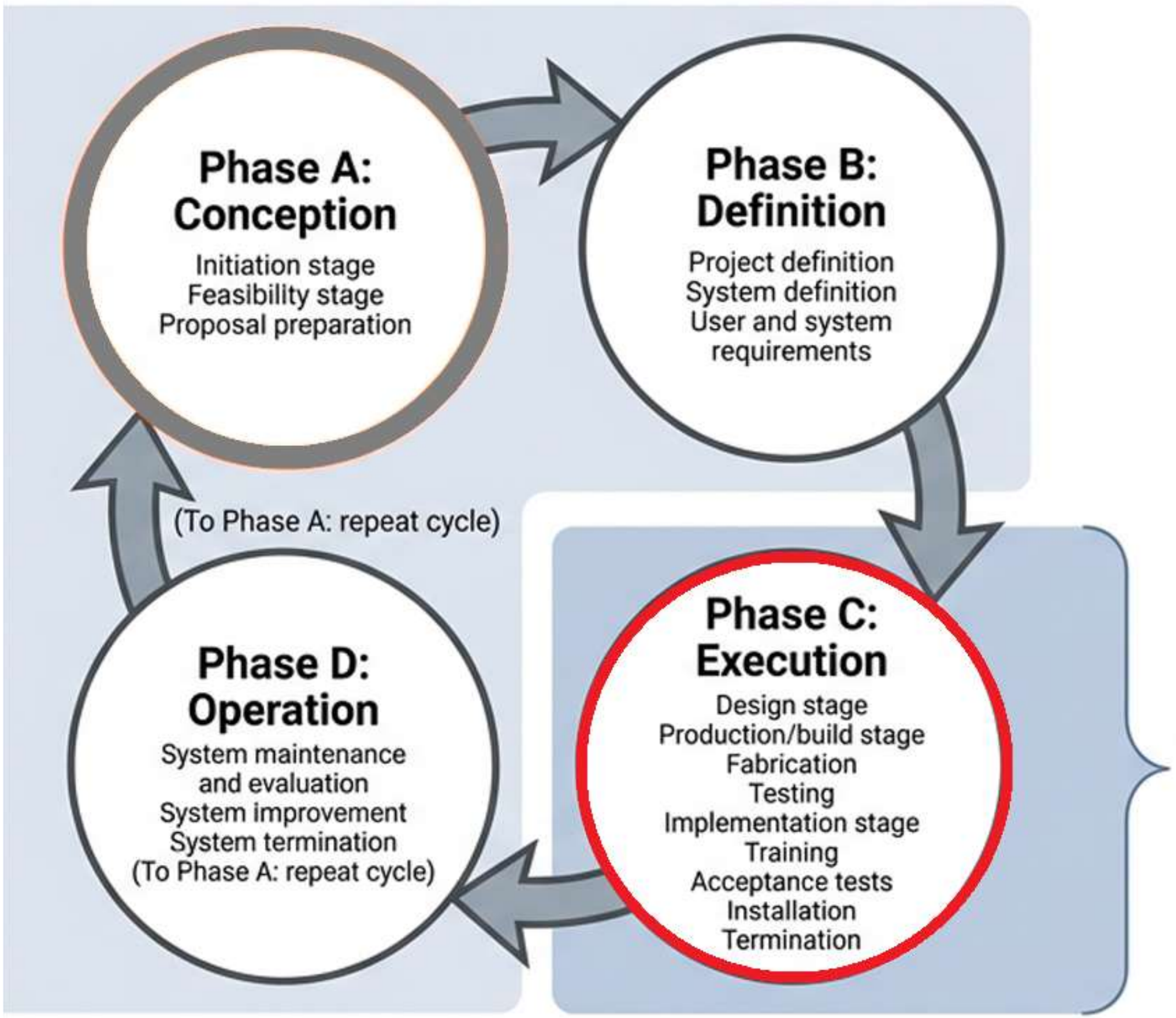
Apply Earned Value Management.

Analyze project performance metrics.

Manage scope and issue logs.

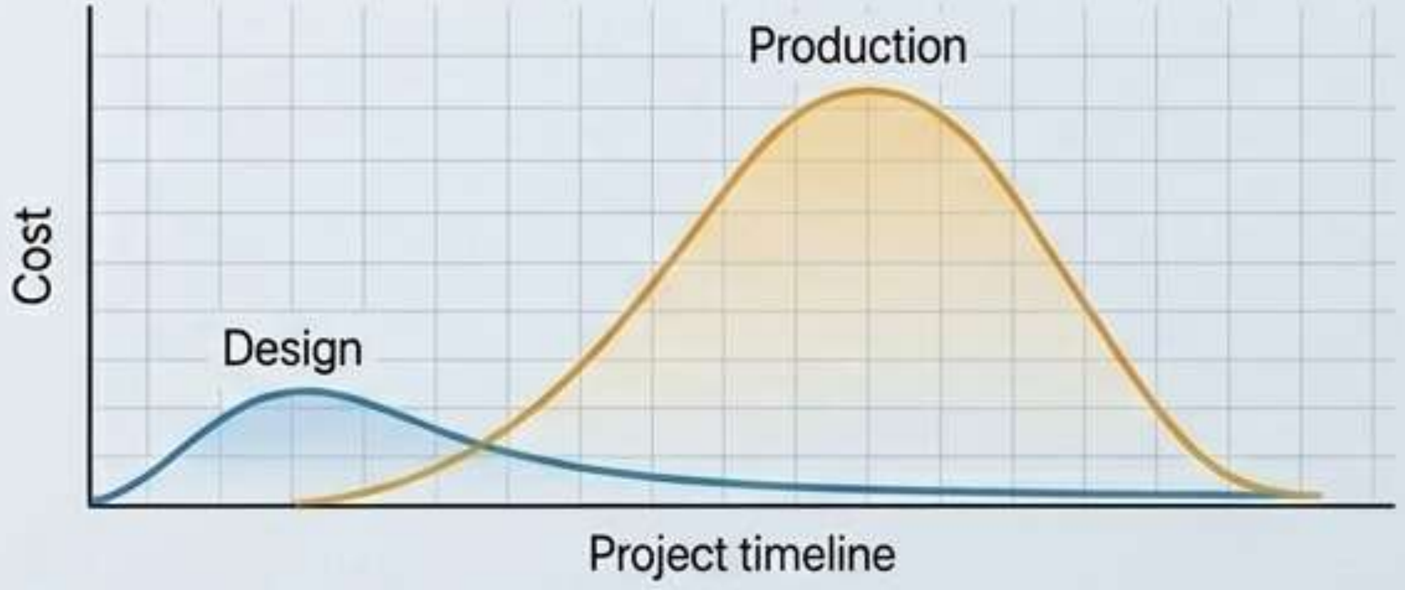
Evaluate overall project progress.

The Systems Development Cycle (SDC)



The project team builds the system (A-C).
The user lives with the system (D).

The Execution Phase



KEY ACTIVITIES & STAGES

Transforms project plan into reality.

Includes detail system design stage.

Includes production and build stage.

OBJECTIVES & IMPACT

Focuses on fabrication and testing.

Prepares system for final delivery.

Consumes largest share of budget.

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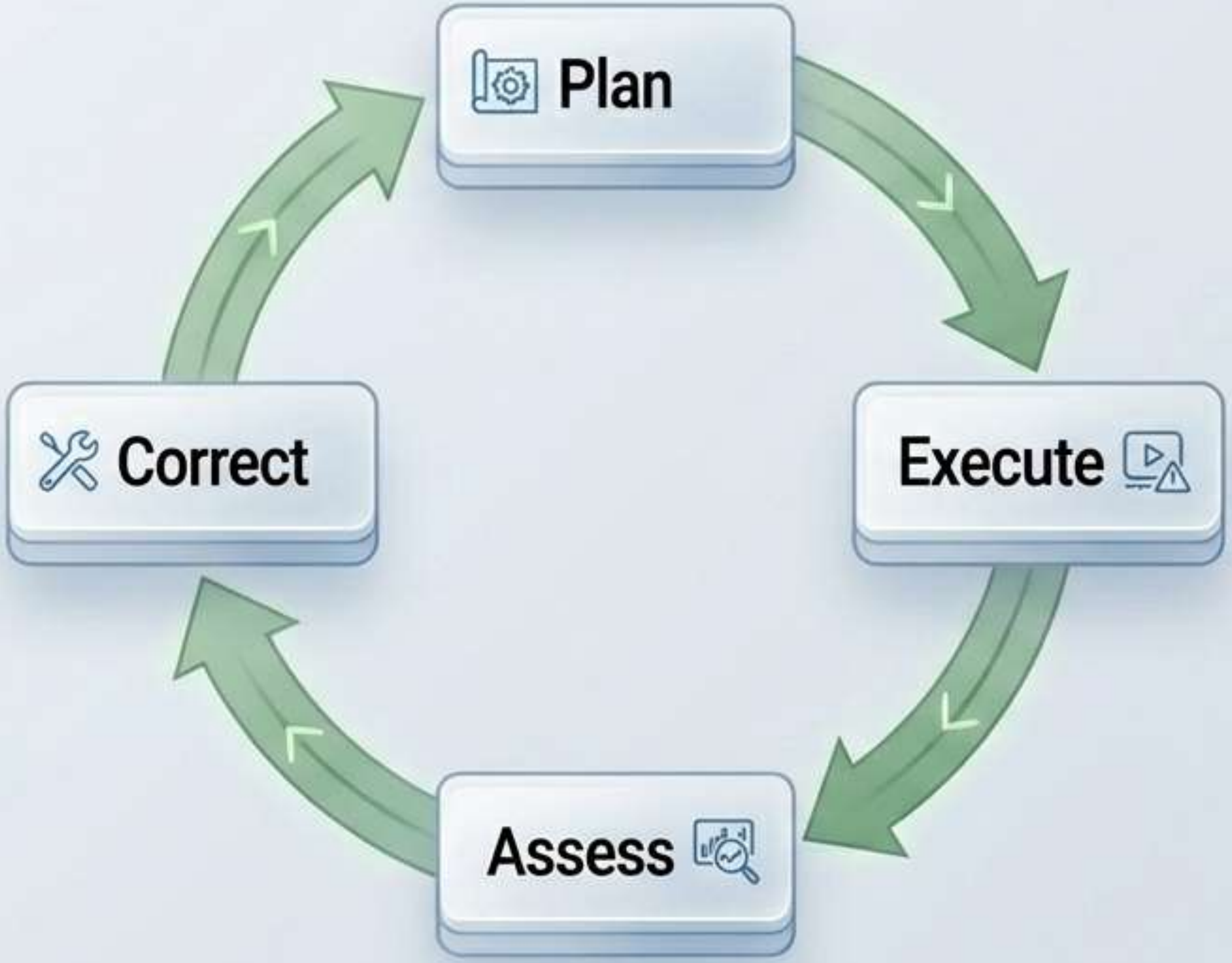
Think of the QuickBite app.

Identify one task per stage:

1. Detail Design stage task?
2. Production or Build stage?
3. Implementation and launch stage?

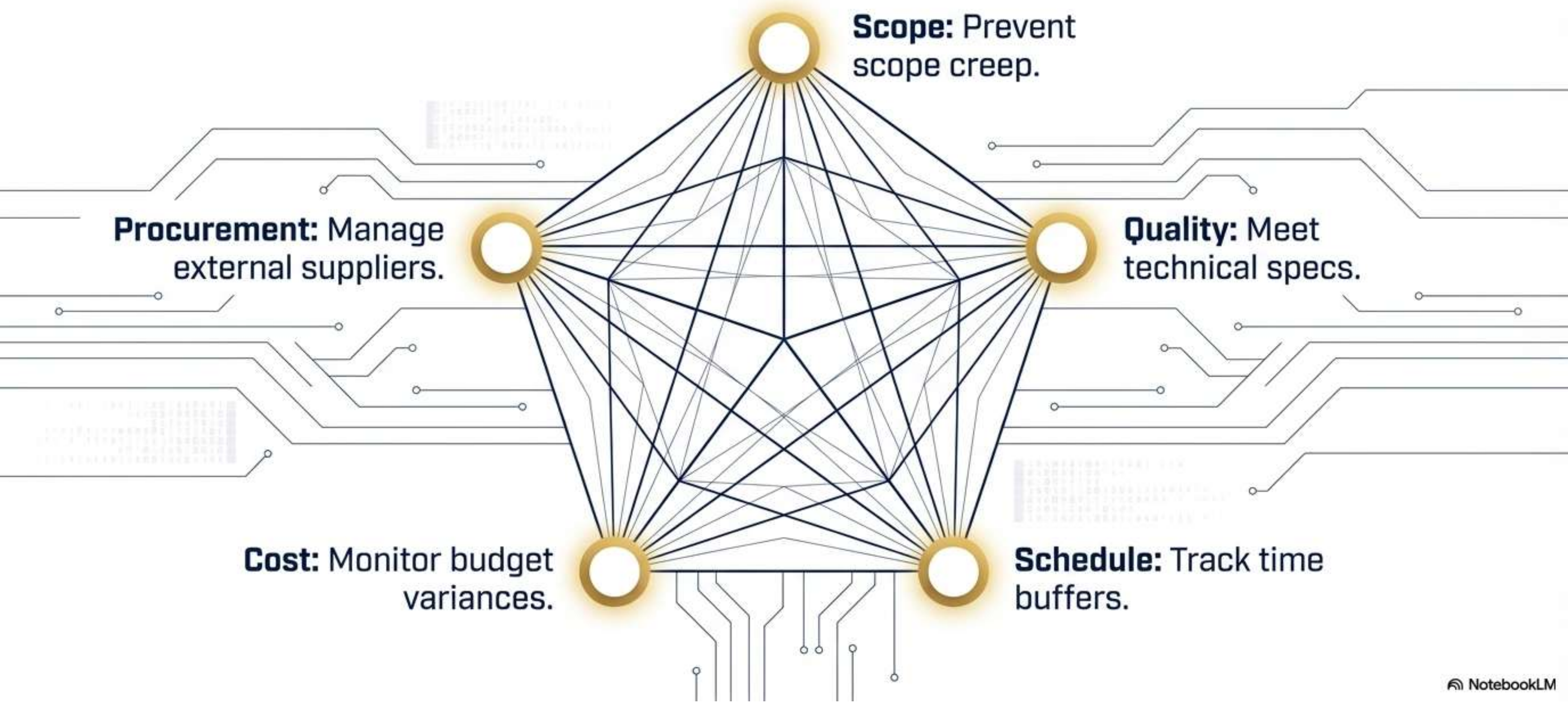
Discuss with your neighbor now.

Monitoring & Control



- Keeps project on targeted path.
- Tracks actual progress versus plan.
- Identifies project deviations very early.
- Takes prompt corrective actions quickly.
- Occurs throughout project life cycle.
- Uses internal and external controls.

Five Control Dimensions



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Practice 1: Spot Deviations

- Read the short scenario.
- Identify the failing dimension.
- Propose immediate corrective action.
- Discuss with your partner.

The QuickBite payment system fails testing, but the timeline remains untouched.

03:00

Tracking Work Progress

Direct supervision and observation.

Monitor task milestone completions.

Conduct tests and demonstrations.

Review technical performance metrics.

Update the issue log.

Earned Value Management

PV: Budgeted planned work.

AC: Actual costs incurred.

EV: Work actually completed.

EV links cost, schedule.

Earned Value Management (EVM) Basics

PV
Planned Value or budget.

EV
Earned Value, work done.

AC
Actual Cost, money spent.

SV: Schedule Variance
(EV minus PV).

CV: Cost Variance
(EV minus AC).

Positive variances mean good performance.

Key EVM Metrics

	Cost	Schedule
Variances (-)	**Cost Variance:** EV minus AC.	**Schedule Variance:** EV minus PV.
Indices (/)	**CPI:** EV divided by AC.	**SPI:** EV divided by PV.

Index under 1 equals trouble.

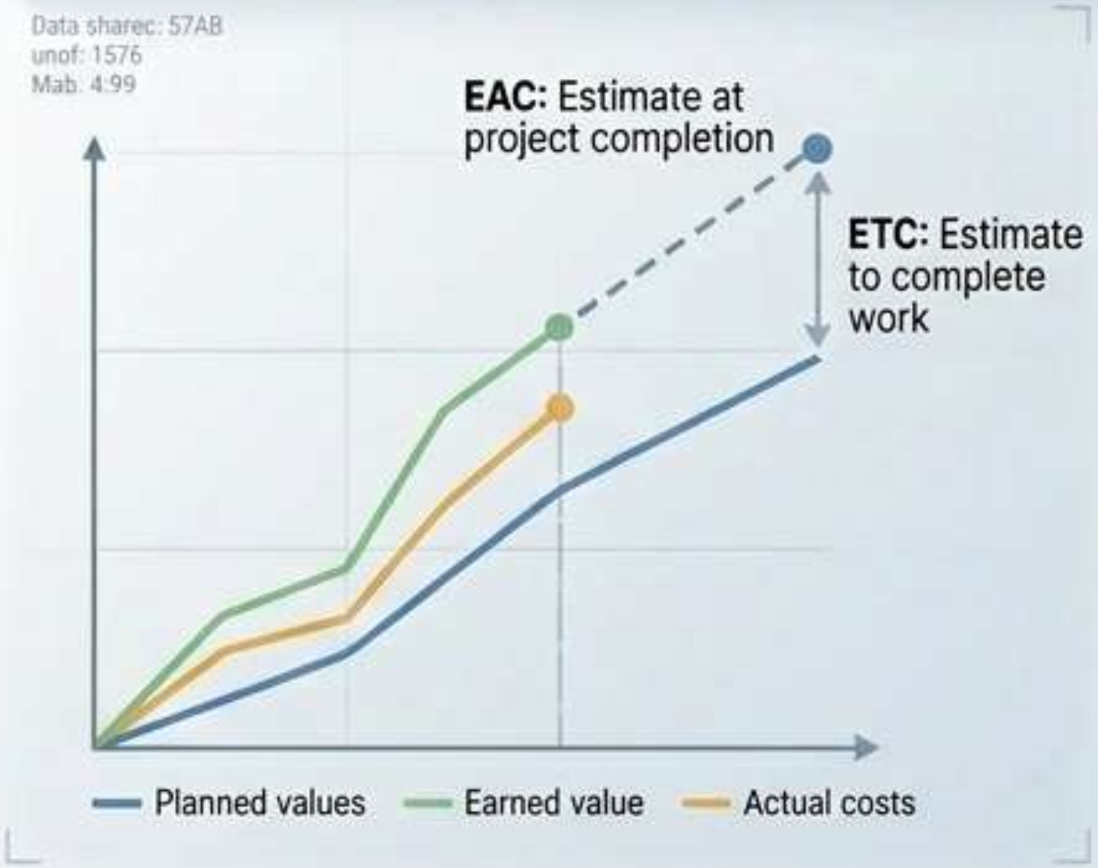
Indices and Forecasting

Current Performance

SPI: Schedule Performance Index (EV/PV)

CPI: Cost Performance Index (EV/AC)

Future Forecasts



Index over one means good.

Forecasts help adjust project plans.

Client requests a new feature.

Just add a quick chatbot.

As manager, what is next?

1. Say yes immediately.
2. Say no immediately.
3. Evaluate schedule and cost impact.

Traditional Tracking

- Traditional compares budget to actuals.
- It ignores actual work completed.
- Matching budget spending is misleading.

EVM Tracking

- EVM integrates actual work performed.
- EVM measures true project health.
- EVM is objective and accurate.

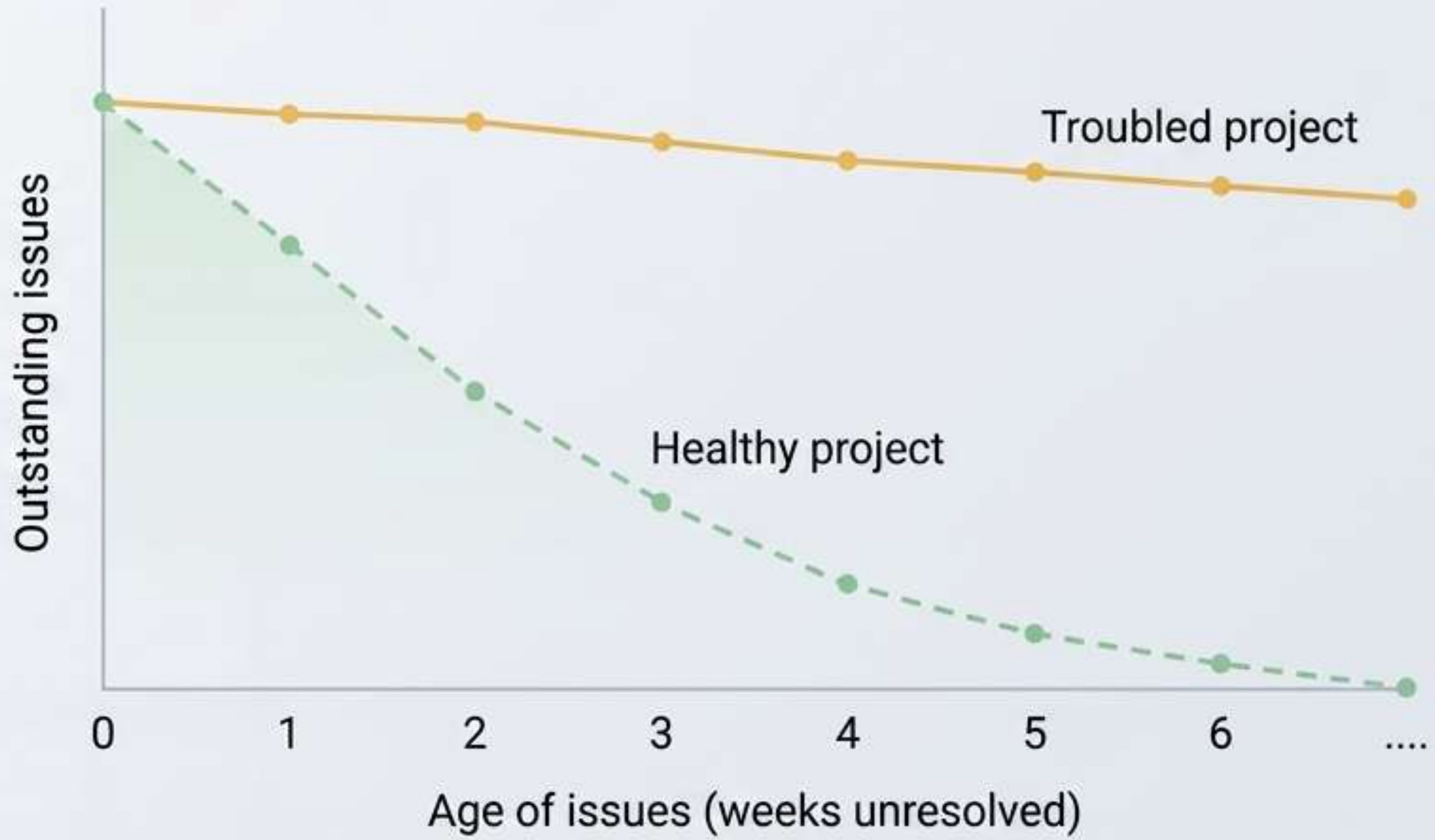
Practice 2: Calculate EVM

- **PV** equals $10,000.
- **AC** equals $12,000.
- **EV** equals $8,000.
- Find CV and SV.
- Find CPI and SPI.
- Is project on track?

CV = -4,000
SV = -2,000
CPI = 0.66
SPI = 0.8

[!] Over budget, behind schedule

Managing Project Issues



- Issues are unresolved project problems.
- Maintain a detailed issue log.
- Track issue age and status.
- Healthy projects resolve issues fast.
- Troubled projects have growing backlogs.
- Prioritize, assign, and close issues.

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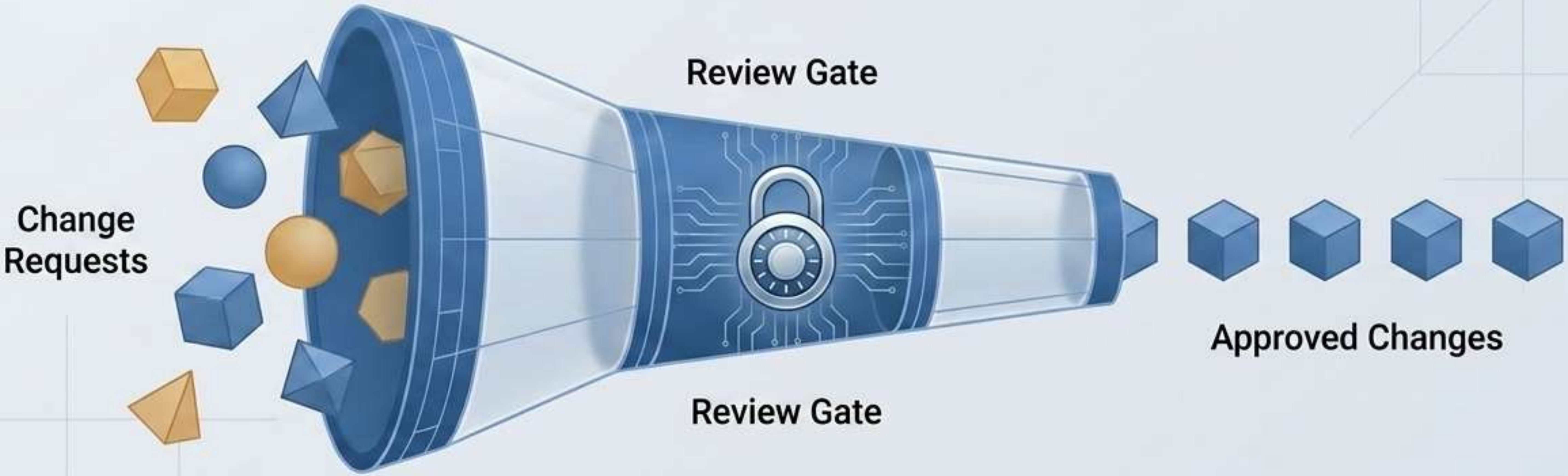
Change Control



- Changes to plan are inevitable.
- Document all new change requests.
- Assess impact on time, cost.

- Use formal change control system.
- Prevent unchecked project scope creep.
- Update baselines after approving change.

Practice 3: Mid-Project Crisis

Client demands new feature.

This is scope creep.

Lead designer falls ill.

This is resource loss.

How do you respond?
Draft a change request.

Key Takeaways

- Execution builds the designed system.
- Control loops correct project deviations.
- EVM integrates cost and schedule.
- Variance signals need for action.
- Issues require logs and tracking.
- Change control prevents scope creep.

Final Assessment

QUESTION

How does EV differ from PV?

WARNING

Why monitor the critical path?

OPERATION

What defines a project issue?

COMMAND BAR

Discuss answers with your group. Prepare to present your findings.